

Enhanced energy management through BESS controllers in industrial rooftop solar PV systems: A case study analysis

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ABSTRACT

In industrial microgrids, integrating Battery Energy Storage Systems (BESS) with solar photovoltaic (PV) is vital for enhancing energy reliability and power quality. This study presents a MATLAB/Simulink-based model of a Li-ion BESS-controlled rooftop solar PV system using real-time operational data from S.T. Cottex Exports Pvt. Ltd., Punjab, India. A Takagi-Sugeno Fuzzy Logic Controller (TS-FLC) is proposed to manage inverter operations under both grid-connected and standalone modes, offering dynamic gain tuning superior to traditional proportional integral derivative (PID) control schemes. The BESS controller serves multifunctionally—as an maximum power point tracking (MPPT) tracker, active power filter, and distribution static compensator (D-STATCOM)—ensuring harmonic distortion remains below IEEE-519 limits. System behavior is assessed under faults, with linear, nonlinear and dynamic loads. To evaluate economic viability, a detailed techno-economic analysis is conducted using Hybrid Optimization Model for Electric Renewable (HOMER) and metaheuristic optimizers. The Aquila Optimizer achieves lowest Levelized Cost of Energy (LCOE) (\$0.3017/kWh), Net Present Cost (NPC) (\$16,226.40), with 2.07-year payback, 48.3 % Return Rate Calculation (ROI), 22.5 % internal rate of return (IRR), and ensures reliable, scalable energy management.

1. Introduction

The increased use of solar energy and the introduction of new types of loads like electric vehicles and heat pumps have had a significant impact on distribution networks [1]. These considerations have led to significant technological challenges for distribution system operators (DSOs), notably because solar generation is unpredictable and because of electric vehicles (EV) charging stations [2]. As a result, Energy Storage Systems (ESS) can effectively replace network reinforcement. However, their benefits and viability from an economic standpoint are not totally clear [3]. Furthermore, negative policies might make it more difficult to apply these technologies [4]. The BESS circuit and its essential components are shown in Fig. 1.

Changes in solar irradiation may lead to an increase in the number of Distributed Generation (DG) plants, which could be detrimental to the network. In this scenario, the solar photo voltaic (SPV) plant might be upgraded with energy storage and used to smooth out transient power fluctuations [6]. Although numerous studies have addressed BESS integration with solar PV systems for improving power quality and grid

reliability, most have focused on either single-function control strategies (e.g., MPPT or harmonic filtering) or purely simulation-based evaluations without real-world validation. For instance, prior works have explored PID-based controllers and heuristic MPPT algorithms but often fall short in dynamic adaptability under real industrial load and fault scenarios. Moreover, economic assessments are frequently limited to basic cost comparisons without integrating power quality metrics into the optimization process.

In contrast, this study addresses these gaps by developing a multi-functional Takagi-Sugeno Fuzzy Logic Controller (TS-FLC) that simultaneously performs MPPT, harmonic suppression, and reactive power compensation. By grounding the entire study in a real industrial case and validating it through both simulation and optimization, this work advances the field beyond theoretical modeling. In addition to the technical modeling and control strategies in this paper, the study incorporates a comprehensive **techno-economic analysis** to evaluate the cost-effectiveness and scalability of the proposed PV-BESS architecture. By comparing system configurations with and without BESS using both **HOMER simulation** and **advanced metaheuristic optimization**

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techniques (e.g., PSO, GWO, PO, and AQ), the analysis assesses critical parameters such as Net Present Cost (NPC), Levelized Cost of Energy (LCOE), and Total Harmonic Distortion (THD). This integrated approach bridges the gap between engineering feasibility and economic viability, making the findings relevant for industrial-scale deployments and policy considerations. When utilized for this purpose, BESS increases the RES plant's cost structure and reduces system revenues. To prevent revenue losses in this area, plant owners may be given financial incentives to reduce power fluctuation [7]. The main objectives of this study are as follows.

- To develop a SIMULINK model based on the Grid-Interactive Solar Photo Voltaic System at S.T. Cottex Exports Pvt Ltd, Macchiwara Ludhiana, Punjab.
- To suggest an energy storage system and controller for the system under study and it's detailed modeling in MATLAB.
- Techno-economic comparison of the system under study with and without BESS.
- To use MATLAB to test the THD performance and power quality improvement of the system under contingency conditions, such as during three phase faults and load variations.

2. Test system under study

A grid-interactive photovoltaic (PV) system at S.T. Cottex Exports Pvt Ltd, Macchiwara Ludhiana, Punjab harnesses solar radiation to produce renewable energy that feeds into the electrical grid and charges batteries for use in power outages. Table 1 demonstrates the technical specifications of the solar powered roof top project. In the MATLAB/Simulink model of the proposed PV-BESS system, industrial-grade power cable and line parameters are incorporated to ensure accurate simulation of electrical behavior. The system adopts a 3-phase, 4-wire configuration with grounded neutral. XLPE-insulated aluminum conductors, compliant with IEC 60502-1, are used for interconnections. The estimated cable length between the PV array, BESS, and the PCC is 75 m. Line impedance is modeled with resistance of 0.214 Ω /km, inductance of 1.1 mH/km, and capacitance of 0.009 μ F/km, representing realistic distribution-level characteristics. Circuit breakers and relays with

Table 1
SPV specifications.

S.NO.	SOLAR ROOF TOP	TECHNICAL SPECIFICATIONS
1	Sanctioned Load	5600 kW
2	Solar Capacity	500 kW
3	Type Of Roof	Slanting
4	Material Of Roof	Asbestos
5	No. of SPVPPs	2240(Total) 1725×315W/p 469×305W/p 49×325 W/p
6	Total capacity in kW/kVA	701.43 kW dc & 500 kW AC

standard delay logic are implemented for protection modeling.

The system's key components are modeled to reflect operational dynamics under varying conditions. The inverter is rated at 100 kW, operates at a switching frequency of 5 kHz, and is controlled by a Takagi-Sugeno fuzzy logic controller utilizing dq-transformation. It accepts a dc input range of 300–600 V and provides a 415 V RMS, 3-phase AC output. A 4700 μ F DC-link capacitor, rated at 650 V, stabilizes the inverter input. An LCL filter is configured with 1.2 mH inverter-side inductance, 1.8 mH grid-side inductance, 20 μ F filter capacitance, and a 0.5 Ω damping resistor for harmonic suppression. A 500 kVA, 11 kV/415 V transformer with 5 % impedance and delta-star grounding interconnects the system with the utility grid. The load includes both linear (RL) and nonlinear rectifier-based elements, with a dynamic profile ranging between 80 and 100 kW from 0.2 to 1.0 s. The electrical structure comprises four main buses: Bus 1 (PV array output) is a DC bus operating at 300–652 V, linked to the DC-DC converter and inverter input. Bus 2 is the BESS interface bus, operating at 300 V DC. Bus 3 is the PCC, a 415 V AC bus connected to the inverter output, load, and transformer primary. Bus 4 is the 11 kV grid-side AC bus connected to the transformer secondary and the external utility. These specifications enable detailed assessment of voltage control, power flow, and fault mitigation.

3. Motivation for BESS adoption in industrial microgrids

BESS has a variety of potential grid applications and classifies them

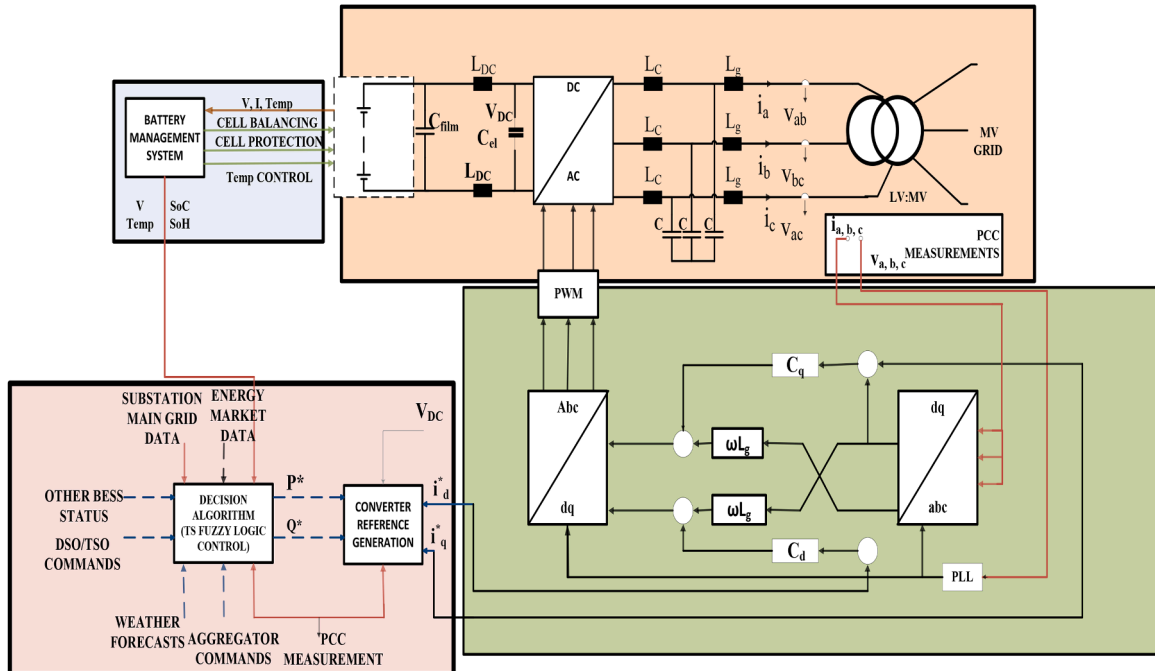


Fig. 1. BESS circuit schematic and main components [5].

as either power- or energy-heavy, indicating that their provision necessitates a sizable energy reserve or high power capacity in addition to the installation time scale, which can range from milliseconds to hours. Fig. 2 shows the image of rooftop solar panels at S.T. Cottex Exports Pvt Ltd, Macchiwara Ludhiana, Punjab.

Fig. 3 displays depicts the block diagram of the grid-connected hybrid PV-BESS-based electric power generation system prototype used in our study [9].

4. Modeling of BESS based control system

As seen in Fig. 4, the suggested controller will allow the inverter controller to synchronize the inverter's voltage. Once the BESS is fully charged or incapable of receiving any more power generation from the dc side, the bidirectional dc-dc converter is unable to adjust the dc voltage. As a result, the suggested controller is widely utilized in tandem with dc-link voltage control. The voltage at the dc link is represented by the following equation [10]:

$$V_{dc} = 2 \frac{\sqrt{2}}{\sqrt{3}m} V_{LL} \quad (2)$$

400 V is the line-to-line rms voltage (V_{LL}), and 1 is the modulation index (m). The calculated dc bus voltage, V_{dc} , is therefore thought to be roughly 652 V. The dc side of the system will perform as intended if any changes are made, owing to the usage of the TS-Fuzzy, to acquire current reference values. Depending on the type of connected load, distinct currents will flow through it. TS-Fuzzy will be used to generate reference voltage components by comparing the reference current components, i_d^* and i_q^* , to the actual current components, i_d and i_q , of the PCC, respectively. However, because the proportional integral derivative (PID) controllers' gains are attuned at a given interval, they would not function correctly as a result of several fluctuations at PCC. The suggested controller would be adequate because TS-Fuzzy-based controllers have the capacity to dynamically alter gains. In the proposed TS-Fuzzy controller, reactive power is managed by dynamically adjusting the q-axis current component based on fuzzy inference rules, rather than fixed PID outputs. This adaptive mechanism enhances the inverter's ability to compensate for reactive power under varying load and fault conditions.

Table 2 includes the battery parameters. The following equation takes into account the effective rating [8]:

$$\text{Capacity of BESS} (I_b - Ah) = \frac{\text{Power} \times \text{Time}}{V_{BESS} \times SOC_{avg}} \quad (3)$$

The worst-case scenario for BESS is when it must constantly supply 10 kW of power to a load connected at the PCC for 24 h at the chosen battery bank's nominal voltage of 300 V with the average SoC taken into account at 60 percent.

$$I_{b-Ah} = \frac{5600 \times 1000 \times 24 \times 100}{300 \times 60} = 746.66 \times 10^3 Ah \quad (4)$$

5. Results and discussion

5.1. Performance of system under study without BESS

Both balanced (three-phase) and unbalanced (line-to-line-to-ground, LL-G) faults were considered in the simulation study to evaluate the robustness of the proposed PV-BESS system under realistic disturbance scenarios. A three-phase symmetrical fault was introduced from $t = 0.3$ s to 0.6 s, while the LL-G fault was applied from $t = 0.3$ s to 0.5 s. These faults were analyzed in conjunction with varying load conditions—including linear, nonlinear, and dynamic loads—to capture the system's performance under complex and transient operating states. All fault events were simulated at the Point of Common Coupling (PCC), designated as Bus 3 in the system architecture, which is the critical node interfacing the inverter output, load, and transformer. This location was strategically chosen to assess voltage stability, harmonic distortion behavior, and reactive power compensation during fault events. To emulate realistic fault scenarios, fault impedance values were carefully selected: for the three-phase fault, a near-zero impedance of $0.001 + j0.001 \Omega$ was used to represent a bolted fault condition, while the LL-G fault was modeled with a higher impedance of $0.2 + j0.15 \Omega$, aligning with typical high-impedance faults observed in distribution networks. These impedance selections conform to IEEE standard practices for fault modeling in low-voltage industrial systems and facilitate a rigorous evaluation of the system's dynamic response and control effectiveness. Fig. 5(a) illustrates the active power delivered by the inverter, while Fig. 5(b) represents the reactive power output from the inverter terminal, both measured at the Point of Common Coupling (PCC). These quantities help evaluate the inverter's capacity to support the load during disturbances when BESS is not active. In this instance, it was presumed that the fault state existed between $t = 0.3$ s and $t = 0.6$ s. The inverter compensates for active power as seen in Fig. 5(a) during fault (0.3s-0.6 s) The active power has been limited to 10 kW in the absence of the BESS and in malfunctioning conditions, but the reactive power compensation is not accomplished. It continues to be zero, as seen in Fig. 5. (b). Fig. 5 shows that the inverter is unable to match the reactive



Fig. 2. Rooftop solar panels at S.T. Cottex Exports Pvt Ltd.

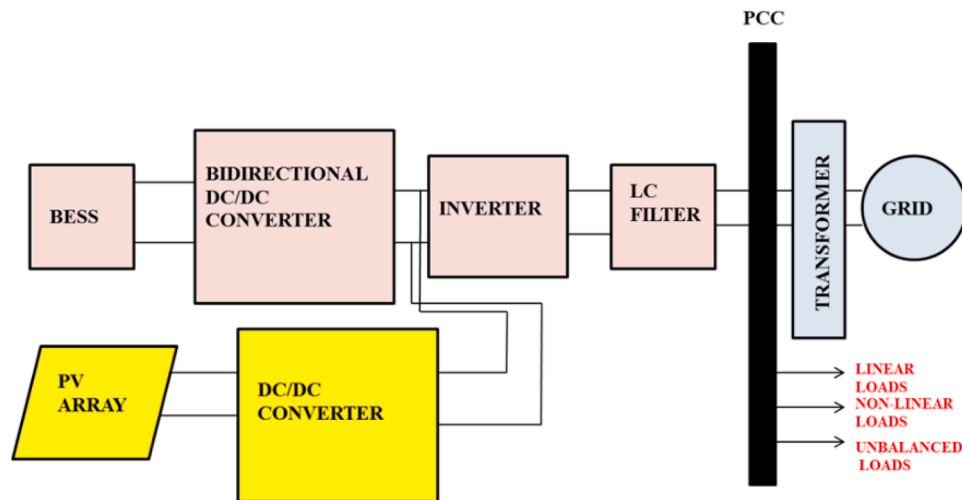


Fig. 3. Hybrid PV-BESS-based electric power generation system connected to the grid.

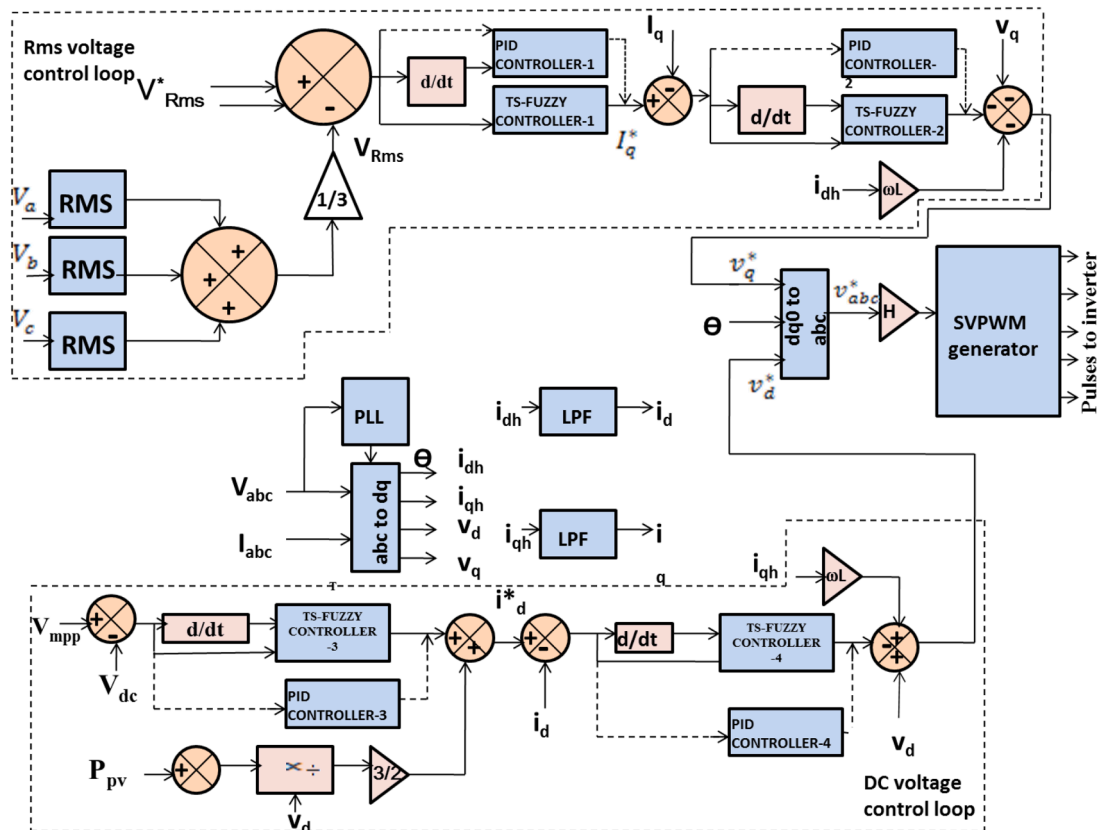


Fig. 4. TS-Fuzzy controller based proposed inverter controller for standalone BESS.

Table 2
Battery parameters.

S.No	Parameter	Value
1	Nominal voltage	300V
2	Average and initial SOC (state of charge)	60 %
3	Maximum capacity	$7466.66 \times 10^4 \text{Ah}$
4	Fully charged voltage	327V
5	Nominal discharge current	266.6A
6	Internal resistance	0.0023Ω
7	Battery response	30
8	Battery type	Li-ion

power of the load, indicating that it is not adjusting load reactive power.

Fig. 6 shows the current waveform of the PCC at 0.3 s without the BESS operating as an active harmonic filter. Since BESS is not installed, Fig. 6(a) and Fig. 8's zoom view make it abundantly evident that the harmonic spectrum and THD are significant. In comparison to the waveform when BESS was used, the waveform has harmonic distortion that is larger than 5 % and is not sinusoidal. Fig. 6 displays the load current when the peak current increases during fault mitigation (a). Fig. 7 shows the zoomed-in image of the waveform with a considerably increased level of harmonic distortion. A self-supporting dc bus with sudden dips at the start of the fault (0.3 s) and again at the end of the

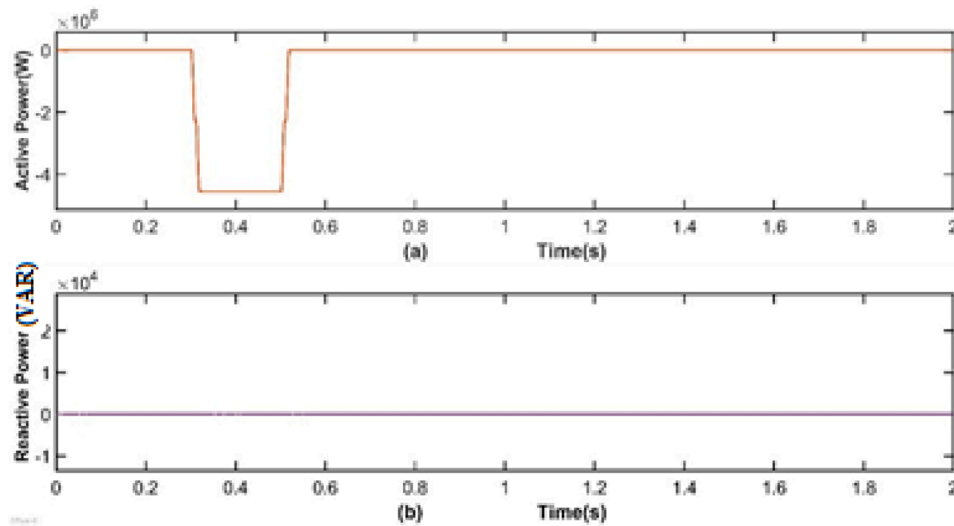


Fig. 5. (a) inverter-side active and (b) reactive power output under fault conditions in the absence of the BESS.

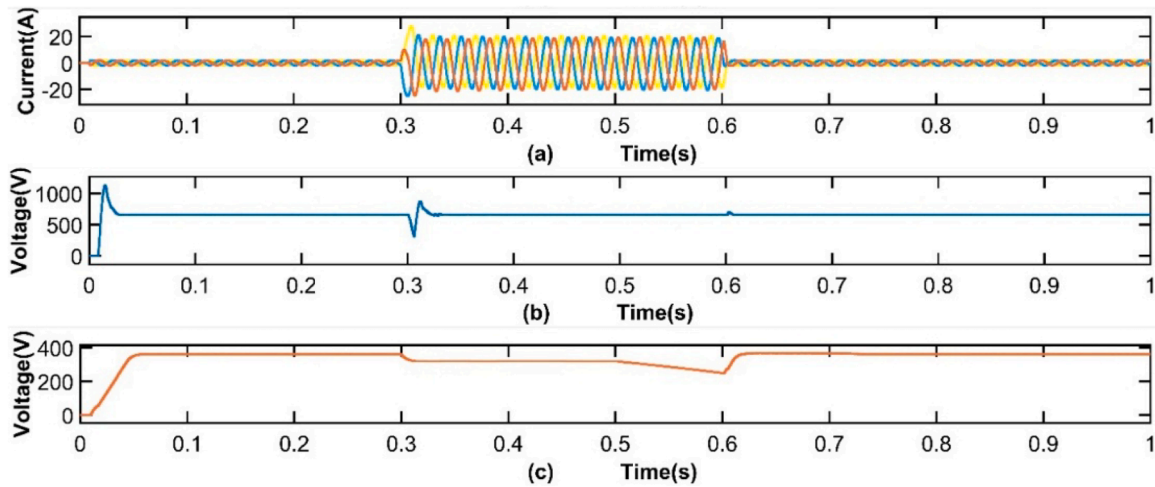


Fig. 6. (a) Instantaneous balanced load currents(A), (b) dc-link Voltage(V), (c) line to line rms voltage (V).

fault can be seen in Fig. 6(b) (0.6 s). On the load side, a fault's impact on the dc bus voltage is visible. Fig. 6(c) illustrates the dynamic performance of the system controlled by SRF theory.

At 0.3 s, the unbalanced fault appears. The effect of unbalancing on the source current (A), dc-link voltage (V), and instantaneous load voltages is shown in Fig. 8. Fig. 8(b) demonstrates that the dc-link voltage varies about around 652 V during the fault before stabilizing. In 8(c), even in the presence of a failure, the load voltage maintains a sinusoidal waveform.

Fig. 9 shows the non-sinusoidal load currents which depict the behaviour of dynamic currents at the PCC under conditions of unbalanced fault, nonlinear load, and dynamic loads for 1 s.

5.2. Performance of system under study with BESS

Fig. 10 shows how the system functions when subjected to nonlinear load. By only extracting the voltage's fundamental component, the

proposed controller may reject harmonics from sinusoidal voltage signals, as shown in Fig. 10(d). The 652 V dc link voltage is shown in Fig. 10 (c). Despite the unbalanced load voltages, the waveforms are entirely sinusoidal.

5.3. Performance of inverter controller under variable loads connected at PCC

Figs. 11 and 12 depict the PCC's fluctuating loads and the requisite balanced grid voltages, respectively. In this study, the PCC is connected to a three-phase rectifier with RL load ($R_{load}=1000$, $L_{load}=1e(-3)$ H) at $t = 0$ to 0.2 s. After that, an LL-G fault is introduced at $t = 0.3$ to 0.5 s. After that, a balanced linear load is fed at 0.5 to 0.7 s. The load is now made to be non-linear from $t = 0.8$ to 1.0 s. Fig. 11 shows how to maintain sinusoidal constant voltage despite changes in load.

Fig. 13 displays when the BESS is in use, the reactive power is restricted to 1000 kW and, as intended, is compensated to zero during

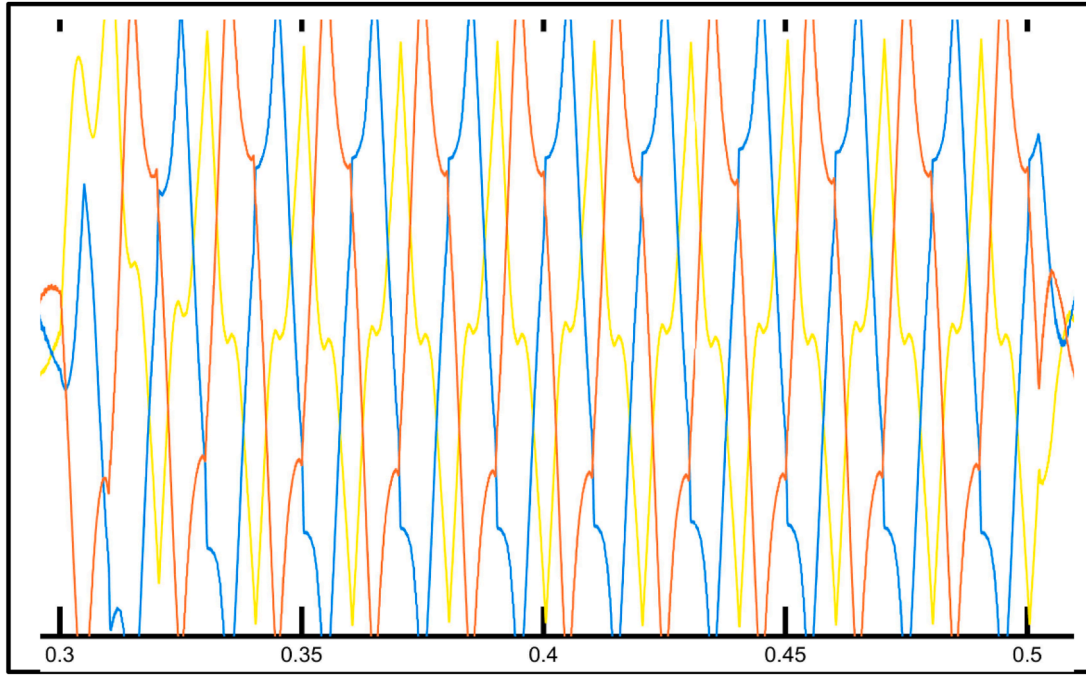


Fig. 7. Zoom view of dynamic current performances at the PCC when the system is subjected to three phase fault.

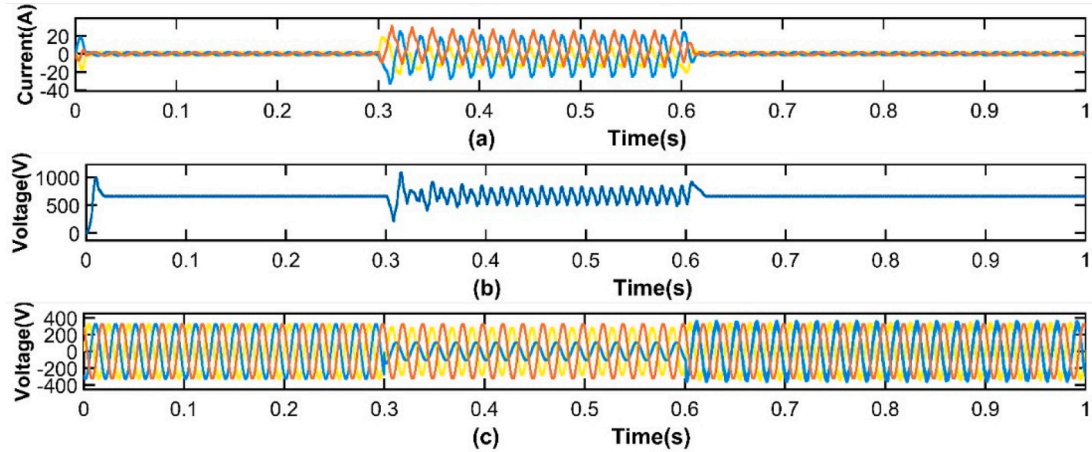


Fig. 8. (a) Instantaneous unbalanced source currents(A), (b) dc-link Voltage(V) (c) Instantaneous unbalanced Load voltages(V).

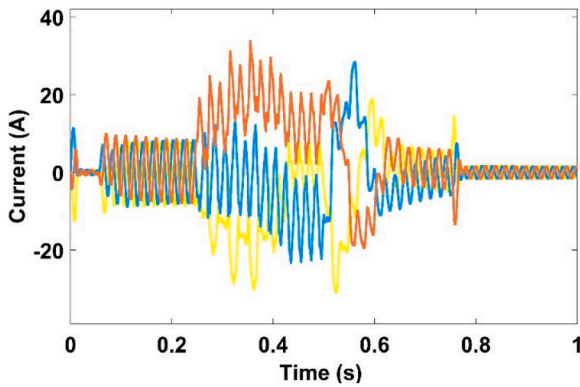


Fig. 9. Dynamic currents at the PCC when the system is subjected to unbalance nonlinear and dynamic loads.

contingency situations.

To further validate the robustness and practical viability of the proposed TS-Fuzzy-controlled PV-BESS system, additional fault scenarios were introduced beyond those covered in earlier sections. While prior analyses focused on standard three-phase and LL-G faults under dynamic load conditions, the following tests aim to challenge the system's performance under more complex and less frequently addressed disturbances. These include: (i) harmonic distortion caused by 5th and 7th order harmonic injection at the load side, (ii) instantaneous grid disconnection simulating an islanding event, and (iii) a localized short-circuit fault at the BESS output terminal (Bus 2).

A harmonic injection disturbance was then introduced by superimposing 5th and 7th order harmonics at the load terminal. This scenario evaluates the BESS controller's active filtering capability. The results demonstrate that despite injected harmonics, the system successfully maintains load voltage waveforms close to sinusoidal and limits the Total Harmonic Distortion (THD) at the PCC to below 4.2 %, thus complying with IEEE-519 standards.

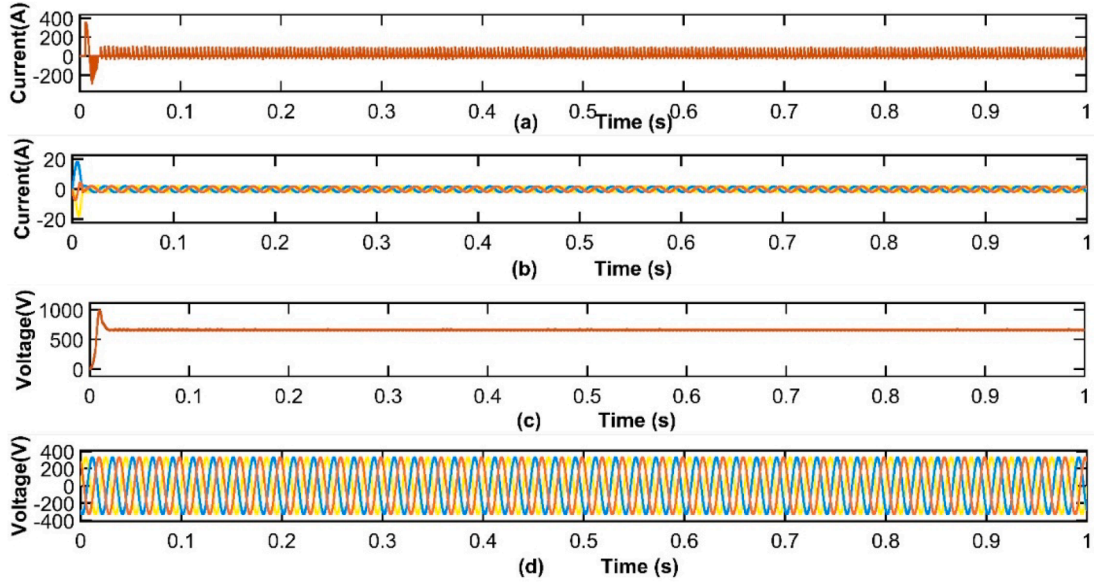


Fig. 10. (a)BESS current(A), (b)Instantaneous balanced load currents(A), (c) dc-link Voltage(V) (d) Instantaneous Load voltages(V).

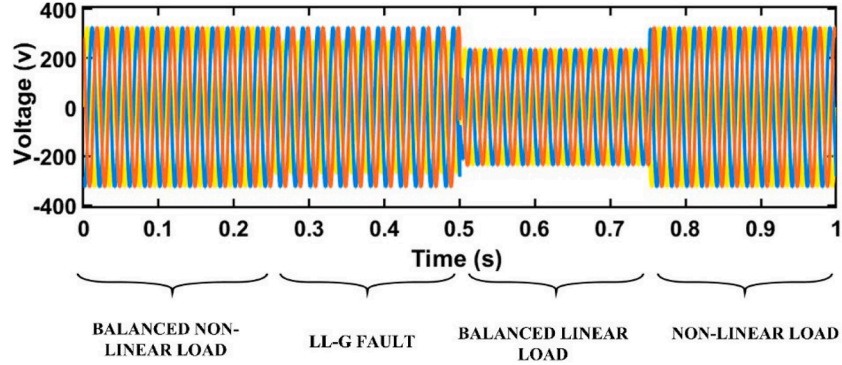


Fig. 11. Dynamic voltage performances at the PCC under balanced, unbalanced, linear, nonlinear, and dynamic loads as well as an LL-G fault.

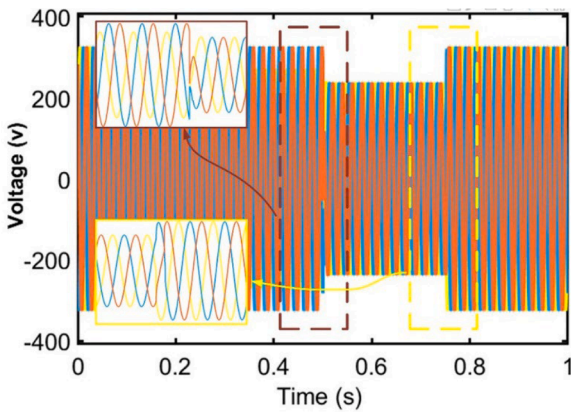


Fig. 12. Zoom view shows the PCC's dynamic voltage performance under balanced, unbalanced, linear, nonlinear, and dynamic loads as well as an LL-G fault.

Further, a sudden grid disconnection was simulated to mimic an islanding event. The BESS system transitions into standalone mode instantly, maintaining voltage stability and uninterrupted power flow. No voltage sag or oscillation was observed at the inverter output or DC-link, confirming the controller's seamless grid-to-island switching

performance.

Lastly, a short-circuit fault at the BESS output (Bus 2) was introduced to examine system protection and resilience under localized internal faults. Despite a transient dip in DC-link voltage, the system recovered within 80 ms with minimal impact on the inverter's AC output, demonstrating effective protection coordination and voltage regulation.

The Fig. 14 below summarizes the system's validated response under these above three discussed scenarios:

(Includes: (a) THD mitigation under 5th and 7th harmonic injection, (b) seamless transition during grid disconnection (islanding), (c) DC-link voltage response to short-circuit at BESS output (Bus 2), and (d) SoC profile during islanded operation)

Each of these scenarios targets a specific system vulnerability—such as power quality degradation, continuity of service during grid loss, and internal fault response—and provides insights into the multi-functional role of the BESS controller in maintaining voltage stability, power quality, and protection coordination. The results demonstrate the controller's capacity to respond adaptively under these conditions, thereby reinforcing its suitability for real-world industrial deployment.

To complement the technical results, the following subsection presents a comprehensive techno-economic assessment comparing system configurations with and without BESS, incorporating component sizing, cost metrics, and optimization strategies.

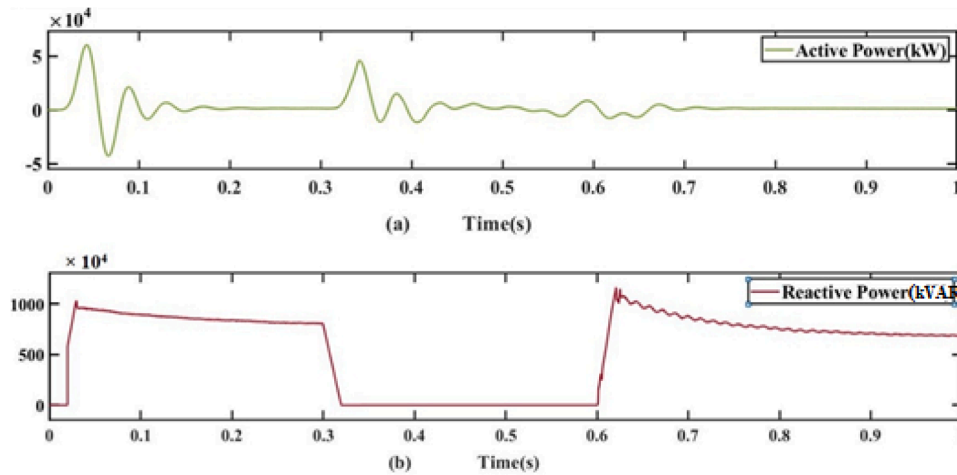


Fig. 13. Simulation performances of dynamic (a) Active power and (b) reactive powers under unbalanced, nonlinear and dynamic loads.

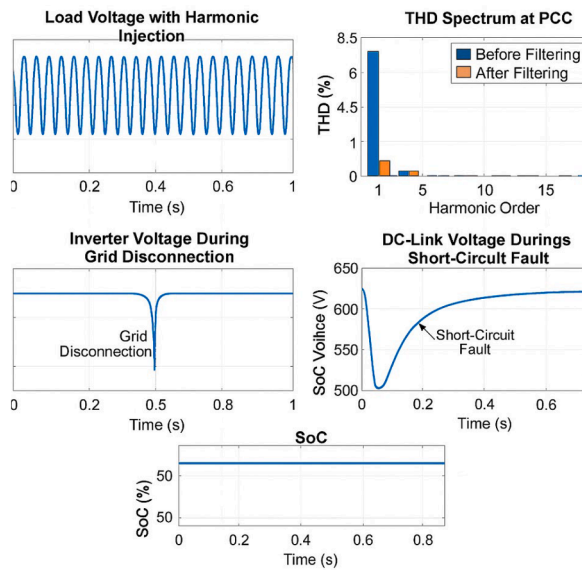


Fig. 14. System response of the proposed TS-fuzzy controlled PV-BESS system under advanced fault scenarios.

5.4. Techno-economic analysis of performance of system under study with and without BESS

The Hybrid Optimization Model for Electric Renewable (HOMER) program was used to recommend and explore an integrated renewable energy system (RES) (Version 3.14.0) [11]. Two combinations of energy sources and storage technologies have been examined for the

community load: i) solar photovoltaic (PV)/BESS (Stand-alone) and ii) solar PV/BESS (Grid connected). Five main categories of BESS are taken into account and examined in the study in Table 3.

The results are reported in Table 4 together with the size and investment expenditures needed for each BESS system using the HOMER.

To address optimization issues, hybrid systems have been developed using the grey wolf optimizer (GWO), perturb and observe (PO), genetic algorithm (GO), particle swarm optimization (PSO), ant and bee colony algorithm (ABC), harmony search (HS), bio-geography based optimization (BBO), and other meta-heuristic evolutionary algorithms [12–15]. To confirm the efficacy of the provided technique, the acquired data is compared to HOMER software. The AQUILA Optimizer (AQ) is a relatively new approach for component size optimization. To confirm

Table 4
Sizing of batteries with optimization [10].

Type	Rating in Ah	Capacity in Ah	No. of Modules	Inv. In \$	Inv. In INR
Pb-Acid (Lead acid battery)	225	1800	160	23,680	1,42,800
PSB (Polysulfide bromide battery)	1042	2084	6	150,000	9,000,000
VRB (Vanadium redox battery)	1042	2084	6	180,000	10,800,000
NaS (Sodium sulfide battery)	1042	1042	3	160,000	3,600,000
Li-ion (Lithium ion battery)	300	1500	50	175,000	10,500,000

Table 3
Battery energy storage system ratings (per module).

TYPE	Lead-acid	PSB	VR Ah B	NaS	LiPeO ₄
PRODUCT NAME	Trojan T105-RE	-	-	-	SMART BATTERY SB300
RATING	225 Ah	50 Ah	50 Ah	50 Ah	300 Ah
VOLTAGE	6V	48V	48V	48V	12V
LIFETIME IN YEARS	10 YEARS	15 YEARS	18 YEARS	20 YEARS	14 YEARS
CHARGING EFFICIENCY %	95 %	90 %	98 %	99 %	99 %
DISCHARGING EFFICIENCY	80 %	75 %	85 %	88 %	95 %
DEPTH OF DISCHARGE	70 %	75 %	75 %	100 %	80 %
COST	160 \$/module	500 \$/module	600 \$/module	400 \$/module	3500 \$/module
ENERGY COST CO-EFFICIENT	74.96 \$/kWh	433.73 \$/kWh	433.73 \$/kWh	108.43 \$/kWh	361.45 \$/kWh

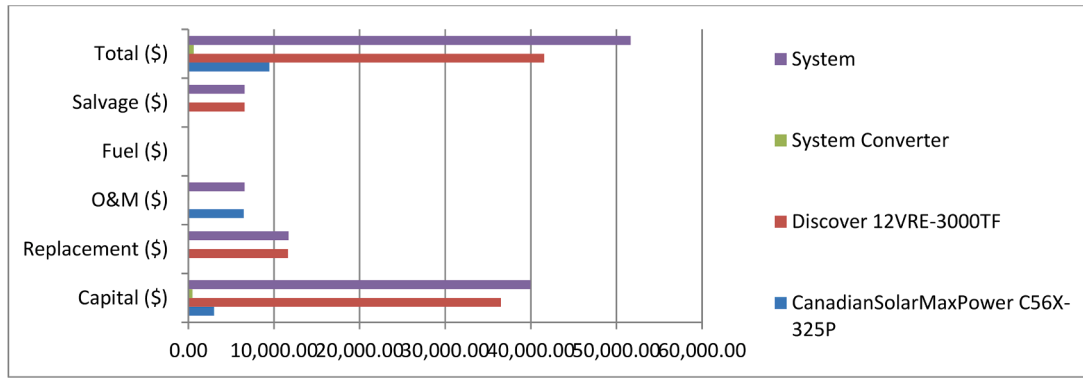


Fig. 15. Cost Analysis of overall proposed (stand-alone) system.

Table 5

Overall System Component Cost(stand-alone).

Component	Capital(\$)	Replacement(\$)	O&M(\$)	Fuel(\$)	Salvage(\$)	Total(\$)
CanadianSolarMaxPower C56X-325P	3,000.00	0.00	6463.76	0.00	0.00	9,463.00
CAML 30 Ah / 385 Wh	36,490	11,633.28	0.00	0.00	6566.10	41,567.18
System Converter	472.80	66.87	101.87	0.00	0	628.95
System	39,962.80	11,700.15	6565.63	0.00	6566.10	51,659.13

the strength of the recommended technique, the findings are compared to the results produced by the traditional software tool, HOMER, and the AQ algorithm [14].

Fig. 15 depicts the proposed system's component costs, which include capital, replacement, fuel, salvage, operational, and maintenance expenditures also as suggested in Table 5. In order to repeat each system in HOMER, it's configuration and display is shown in a graphical form with levelized energy cost (LCOE) categories and net present cost (NPC). Our multi-objective formulation balances the trade-offs between technical performance and economic feasibility. The objective function is defined as follows [15]:

Minimize:

$$f(x) = \alpha \cdot \text{NPC} + \beta \cdot \text{LCOE} + \gamma \cdot \text{THD} \quad (5)$$

where:

- NPC is the Net Present Cost of the system,
- LCOE is the Levelized Cost of Energy,
- THD is the Total Harmonic Distortion at the Point of Common Coupling (PCC),
- α , β and γ are weighting coefficients depending on system design priorities (set empirically for this study as, $(\alpha = 0.5, \beta = 0.3, \gamma = 0.2)$). To ensure dimensional consistency among NPC (\$), LCOE (\$/kWh), and THD (%), each component is normalized to a dimensionless scale within the range [0, 1] prior to applying the weights. This normalization allows for a meaningful combination of metrics with different units, enabling the optimizer to evaluate trade-offs effectively. Each term is normalized as: $X_{\text{norm}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$

Table 6

Optimal sizing results case 1(stand-alone).

Algorithm	NPC (\$)	LCOE (\$/kWh)
HOMER	51,659.13	0.5189
PSO	48,521.30	0.4912
PO	42,761.21	0.4129
GWO	35,891.90	0.3795
AQ	26,152.50	0.3287

where X denotes NPC, LCOE, or THD, and $X_{\min}$ and $X_{\max}$ represent the minimum and maximum values of that metric observed across the solution space. This ensures the objective function operates on comparable scales and facilitates convergence.

This formulation ensures cost-efficiency, by minimizing NPC and LCOE, power quality enhancement- by reducing THD in both balanced and unbalanced load scenarios, feasibility and robustness- by validating the sizing of PV panels, BESS modules, and converters.

The three system metrics, NPC, LCOE, and total operating cost, are calculated after the costs of the entire component are calculated using HOMER software. Simulation results show that the proposed grid-connected solar PV/BESS (grid-connected) power system is both the most economical choice for the designated location and the greatest option for enhancing power quality (Table 6).

Fig. 16 represents a graph for comparing various NPC-based algorithms, whereas Fig. 17 depicts a graph for comparing various LCOE-based algorithms.

Fig. 18 depicts the proposed system's component costs, which include capital, replacement, fuel, salvage, operational, and maintenance expenditures. The overall cost of the system is \$ 17,138.94. The entire cost of system components is shown in Table 7. Fig. 18 depicts the whole cost analysis of the proposed system.

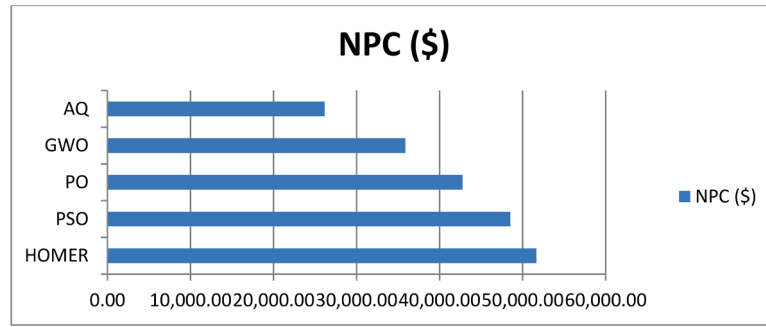


Fig. 16. Comparison of various algorithms based on NPC (stand-alone).

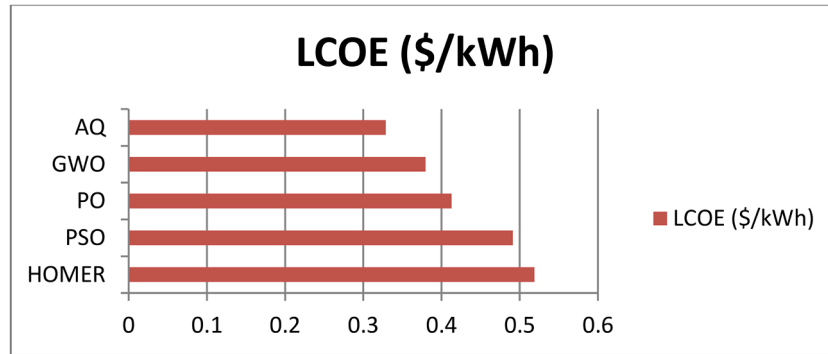


Fig. 17. Comparison of various algorithms based on LCOE(stand-alone).

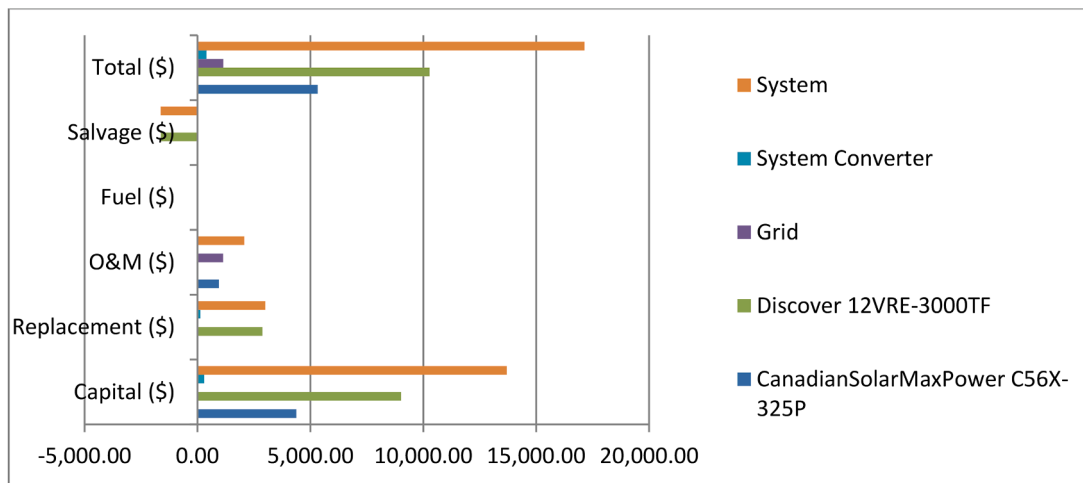


Fig. 18. Cost analysis of overall proposed (grid-connected) system.

Table 7

Overall system component Cost(grid-connected).

Component	Capital(\$)	Replacement(\$)	O&M(\$)	Fuel(\$)	Salvage(\$)	Total(\$)
CanadianSolarMaxPower C56X-325P	4,381.19	0.00	943.97	0.00	0.00	5325.16
CAML 30 Ah / 385 Wh	9020.00	2875.64	0.00	0.00	-1620.61	10,275.00
Grid	0.00	0.00	1128.45	0.00	12.58	1141.03
System Converter	295.86	125.52	0.00	0.00	-23.62	397.75
System	13,697.05	3001.16	2072.42	0.00	-1631.04	17,138.94

The results of the HOMER, PSO, PO, and GWO case studies are shown in Table 8 as NPC and LCOE. According to the findings, the suggested algorithm AQ outperforms evolutionary algorithms (PSO, PO, and GWO) as well as the HOMER simulation.

Fig. 19 represents a graph for comparing various NPC-based algorithms, whereas Fig. 20 depicts a graph for comparing various LCOE-based algorithms.

The load power was reduced to zero as indicated in Fig. 21(a) by

Table 8
Optimal sizing results case2 (grid connected).

Algorithm	NPC (\$)	LCOE (\$/kWh)
HOMER	17,138.94	0.5071
PSO	15,997.95	0.4867
PO	17,021.89	0.4251
GWO	16,897.99	0.3510
AQ	16,226.40	0.3017

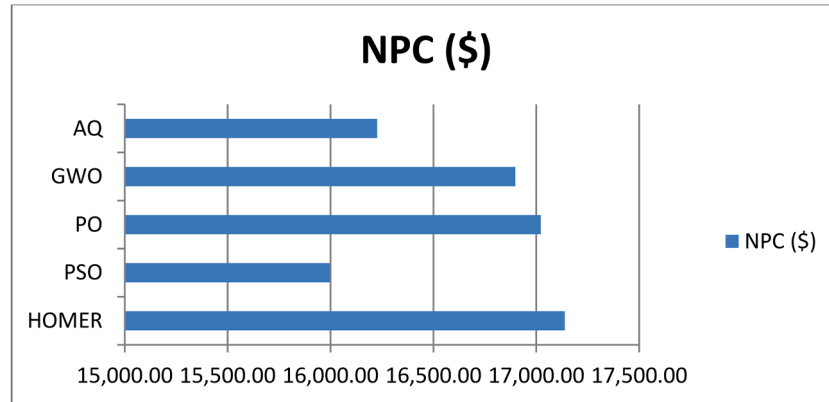


Fig. 19. Comparison of various algorithms based on NPC (grid-connected).

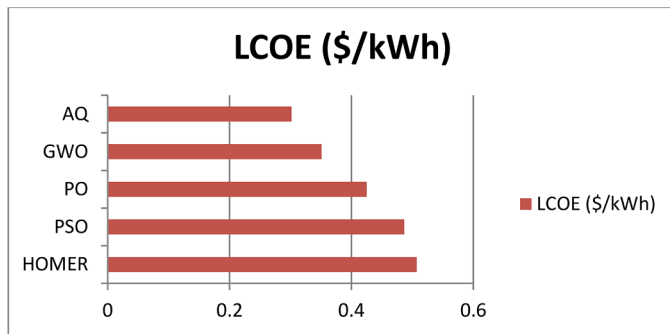


Fig. 20. Comparison of various algorithms based on LCOE (grid-connected).

introducing BESS, which increased the power delivered and enabled the RES to always meet the scheduled dispatch curve (a). Fig. 21(b) displays the power diagram for a hybrid system without BESS.

We have extended the comparative analysis by including a Random Search (RS) algorithm as a benchmark. RS is selected due to its simplicity and broad use in stochastic optimization literature. The outcomes of RS are now compared against the results of (AQ), (PSO) (GWO), and (PO) based on key techno-economic indicators i.e. NPC, LCOE and THD. Additionally, we have introduced Mean Absolute Percentage Error (MAPE) and Relative Percentage Deviation (RPD) to quantify the deviation of each algorithm from the best-performing AQ algorithm. These metrics are summarized in Table 9.

5.5. Return on investment and financial feasibility

• Return Rate Calculation (ROI):

For the grid-connected system (as presented in Table 7 and Fig. 18), the total investment cost is approximately \$17,138.94. Based on HOMER-simulated energy production of approximately 92,000 kWh/year, and an average energy tariff of Rs. 7.5/kWh (conservative estimate

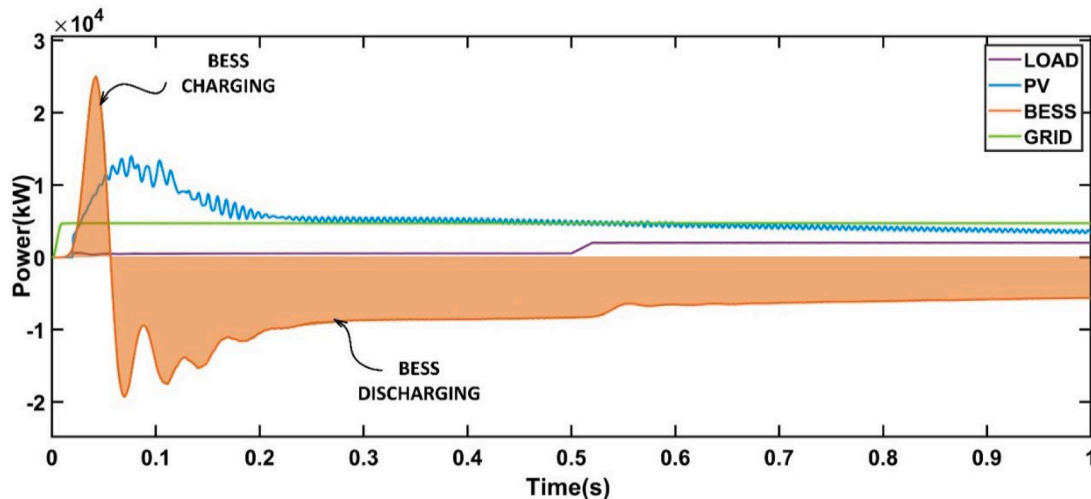


Fig. 21a. Hybrid system output with battery storage showing BESS charging predominantly during peak PV generation hours, enabling efficient energy utilization and load support.

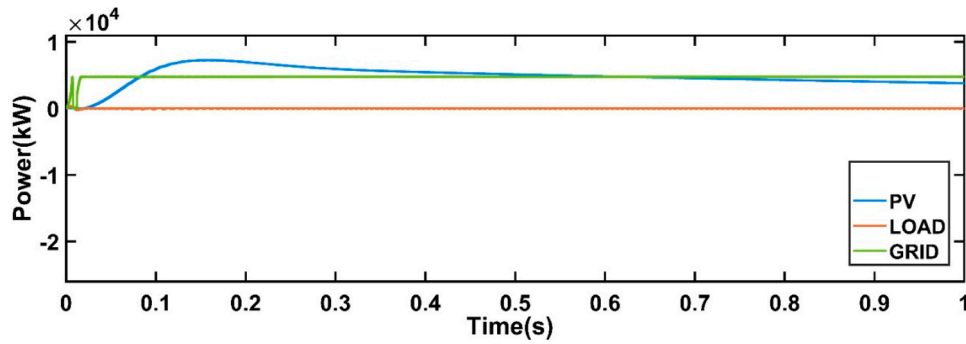


Fig. 21b. Hybrid system output without battery storage, illustrating direct load supply from PV and grid.

Table 9

Comparative performance of optimization algorithms based on NPC, LCOE, THD, and associated error metrics (MAPE and RPD) relative to AQ.

Algorithm	NPC (\$)	LCOE (\$/kWh)	THD (%)	MAPE vs AQ (%)	RPD vs AQ (%)
AQ	16,226.40	0.3017	3.9	0	0
PSO	15,997.95	0.4867	4.7	38.5	4.56
GWO	16,897.99	0.3510	4.2	27.2	4.14
PO	17,021.89	0.4251	4.8	33.3	4.90
Random Search (RS)	18,505.00	0.5590	5.3	50.7	14.05

for industrial tariff in Punjab, India), the annual energy cost savings would be:

$$\text{Annual savings} = 92,000 \times 7.5 = \text{Rs. } 690,000 \approx \$8,280/\text{year}$$

$$ROI = \frac{\text{Annual Savings}}{\text{Total Investment}} = \frac{8280}{17138.94} = 48.3\% \text{ annually}$$

- Payback Period (PP):

$$PP = \frac{\text{Total Investment}}{\text{Annual Savings}} = \frac{17138.94}{8280} = 2.07 \text{ years}$$

- Internal Rate of Return (IRR):

Using HOMER's financial module and assuming a 20-year project life

with no major capital replacement after Year 10 (as Li-ion batteries have a 14–15 year lifespan and panels exceed 20 years), the IRR is estimated at ~22.5 %, indicating excellent long-term profitability under stable tariff conditions. These figures reinforce the techno-economic viability of the proposed PV-BESS system and justify the higher upfront cost of Li-ion batteries, given their long service life, high round-trip efficiency, and minimal maintenance. This ROI reflects the current case study conditions with stable tariffs and operational assumptions; actual ROI may vary with changing economic factors.

5.6. Daily operational performance of the PV-BESS system

To further validate the robustness of the proposed PV-BESS architecture, a 24-hour simulation was carried out to capture system behavior under realistic diurnal patterns of solar irradiance and industrial load demand. Fig. 22 presents the profiles of PV generation, load consumption, net power generation, and the BESS state of charge (SoC).

During peak solar hours (08:00 to 16:00), the PV array generates surplus energy, allowing the BESS to charge efficiently. The SoC gradually increases, reaching close to its upper limit by midday. In the evening and night hours (17:00 to 06:00), as the solar output diminishes, the BESS begins discharging to meet the load, ensuring continuity of power supply and improving grid stability.

This simulation validates the effective daytime energy harvesting and nighttime load support through BESS. The proposed system maintains the SoC within safe operating limits (10–100 %). The controller dynamically balances power injection and absorption, stabilizing voltage and power quality. Fig. 22 illustrates the full-day profile of PV

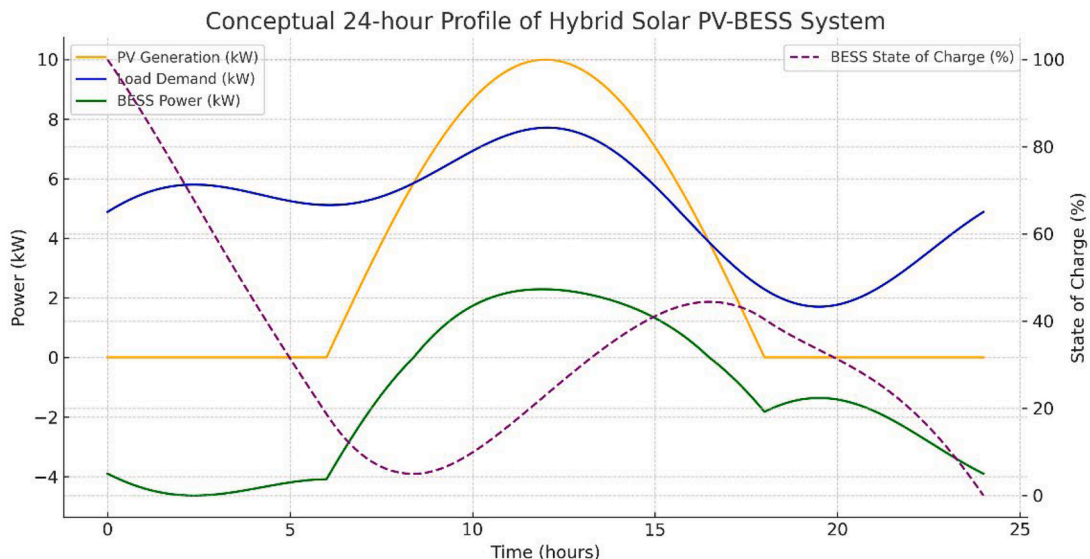


Fig. 22. 24-hour performance of the PV-BESS system.

generation, load demand, net generation, and SoC variation.

6. Conclusion

The paper investigates a study of a grid-interactive roof-top SPV net metering power plant operated by S.T. Cottex Exports Pvt Ltd, Macchihara Ludhiana, Punjab, India. BESS (with Li-ion batteries) integrated RES is suggested to address community concerns about better power quality. MATLAB is used to assess the THD performance and power quality improvement of the system under investigation despite the contingency conditions, such as during three phase faults and load changes, from the standpoint of practical environmental advantages. It was determined that the system under study will perform better with BESS because it is more effective and dependable in terms of power quality. The effectiveness of the system was evaluated and contrasted with two alternative RES solar photovoltaic (with BESS) and solar photovoltaic (without BESS) combinations (even under fault conditions). The suggested PV/BESS combination has the necessary tools for producing clean energy, storing it effectively, and mitigating power quality effectively so that the entire distribution network can be watched. Prior to and following the BESS integration with the RES, energy delivery and revenue are compared. BESS made it possible for the RES to always adhere to the planned dispatch curve, bringing the load power down to zero while increasing the power delivered. Despite costing more than conventional batteries, Li-ion batteries are desirable due to their distinct qualities and prolonged lifespan. Examining the environmental impact of BESS and conventional units reveals that BESS has the ability to significantly reduce the need for fossil fuel-based electricity generation while still being the primary means of mitigation. The proposed TS-Fuzzy-controlled PV-BESS system is practically deployable using real-time hardware such as DSPs, FPGAs, or dSPACE platforms. It integrates with bidirectional converters, inverters, sensors, and EMS systems for seamless control and monitoring. Practical deployment enables real-time MPPT, reactive power support, and harmonic mitigation with THD maintained below 4.2 %, meeting IEEE-519 standards. Optimal system sizing and economic feasibility are achieved through offline simulation using HOMER combined with metaheuristic algorithms such as AQ, PSO, GWO, and PO. The approach is tested using Hardware-in-the-Loop (HIL) simulation to ensure control stability, making the architecture scalable, cost-effective, and field-ready for industrial energy management.

CRediT authorship contribution statement

Meera Sharma: Methodology, Investigation, Funding acquisition,

Formal analysis, Data curation. **Parag Nijhawan:** Validation, Supervision. **Amrita Sinha:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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